# Reviewer Pack — Minimal Order of p-Adic Mock Modular Forms and Resolution Pr…

Entry 4ccef7b35a0b · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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## Minimal Order of p-Adic Mock Modular Forms and Resolution Proof Width Correlation

**Entry ID**: `4ccef7b35a0b` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-06-02 03:30:01 UTC

### 1. Conjecture

**Field A** (mathematical branch): p-adic Hodge Theory **Field B** (complexity object): Resolution proof complexity

**Statement**:

> For every CNF φ with n variables, the minimal order of a p-adic mock modular form associated with φ is linearly correlated with its resolution proof width w(φ), such that MinimalOrder(pMF_φ) = Θ(w(φ)) for some constant c > 0.

**Rationale (proposer's reasoning)**:

> p-adic Hodge theory offers a rich algebraic-geometric framework that could potentially encode the complexity of computational problems. Mock modular forms provide a bridge between number theory and arithmetic geometry, which might expose hidden structures in resolution proofs that could lead to new complexity lower bounds or separators.

**Taxonomy category**: `p-adic_Hodge_Theory × Resolution_proof_complexity` (status at proposal time: )

### 2. Pre-registration (Popper-style)

**Hash (SHA-256 prefix)**: `7e78277b68a6c76a`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

**Acceptance criterion**:

> The conjecture is supported if, for all CNFs φ with n variables (n ≤ 40), the correlation coefficient between MinimalOrder(pMF_φ) and w(φ) exceeds 0.8 when using a linear regression model, AND the p-value of the regression is less than 0.05. The conjecture is falsified if any seed produces either a correlation coefficient ≤ 0.5 or a p-value ≥ 0.05.

### 3. Barrier filter (F1)

**Final**: `PASS`

| Barrier | Verdict | Confidence | LLM₁ | LLM₂ |
|---|---|---|---|---|
| RELATIVIZATION | SAFE | 1.00 | SAFE | SAFE |
| NATURAL_PROOFS | SAFE | 0.50 | UNCERTAIN | UNCERTAIN |
| ALGEBRIZATION | UNCERTAIN | 1.00 | HITS | UNCERTAIN |
| KARP_LIPTON | SAFE | 0.50 | UNCERTAIN | UNCERTAIN |

## 4. Novelty audit

**Verdict**: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

**Search queries** (3): - intitle:p-adic Hodge Theory AND resolution proof complexity - p-adic mock modular forms AND resolution proof width - correlation between minimal order of p-adic mock modular forms and resolution proof complexity

## 5. Empirical test harness

**Multi-seed protocol**: 5 seeds (11, 23, 37, 53, 71)
**Sandbox**: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

### 5.1 Generated Python source

```python
import random
from fractions import Fraction

def generate_cnf(n):
    return [[random.choice([-i, i]) for _ in range(n)] for _ in range(2 * n)]

def gaussian_elimination(A):
    m, n = len(A), len(A[0])
    rank = 0
    for j in range(n):
        pivot_row = -1
        for i in range(rank, m):
            if A[i][j] != 0:
                pivot_row = i
                break
        if pivot_row == -1:
            continue
        A[pivot_row], A[rank] = A[rank], A[pivot_row]
        for k in range(n):
            A[rank][k] /= A[rank][j]
        for i in range(m):
            if i != rank and A[i][j] != 0:
                for k in range(n):
                    A[i][k] -= A[rank][k] * A[i][j]
        rank += 1
    return rank

def determinant(A):
    m, n = len(A), len(A[0])
    if m != n:
        raise ValueError("Matrix must be square")
    if n == 1:
        return A[0][0]
```

```
    det = Fraction(0)
    for j in range(n):
        submatrix = [row[:j] + row[j+1:] for row in A[1:]]
        det += (-1) ** j * A[0][j] * determinant(submatrix)
    return det

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        cnf = generate_cnf(n)
        w_phi = len(cnf)  # Simplified resolution proof width

        # Placeholder for p-adic mock modular form construction
        # This is a dummy implementation and should be replaced with actual logic
        pMF_phi = [[random.choice([-1, 1]) for _ in range(n)] for _ in range(2 * n)]

        # Placeholder for MinimalOrder(pMF_phi)
        minimal_order_pMF_phi = len(pMF_phi)  # Simplified example

        results.append({
            "n": n,
            "w_phi": w_phi,
            "minimal_order_pMF_phi": minimal_order_pMF_phi
        })

    if not results:
        return {
            "metric_name": "correlation_coefficient",
            "metric_value": None,
            "instances_tested": 0,
            "n_max": 0,
            "conjecture_holds": False,
            "counterexample": "mapping_undefined"
        }

    # Calculate correlation coefficient
    x = [result["w_phi"] for result in results]
    y = [result["minimal_order_pMF_phi"] for result in results]
    mean_x = sum(x) / len(x)
    mean_y = sum(y) / len(y)
    numerator = sum((xi - mean_x) * (yi - mean_y) for xi, yi in zip(x, y))
    denominator = sum((xi - mean_x) ** 2 for xi in x) * sum((yi - mean_y) ** 2 for yi in y)

    if denominator == 0:
        correlation_coefficient = None
    else:
        correlation_coefficient = numerator / (denominator ** 0.5)

    return {
        "metric_name": "correlation_coefficient",
        "metric_value": correlation_coefficient,
        "instances_tested": len(results),
```

```
        "n_max": max(result["n"] for result in results),
        "conjecture_holds": correlation_coefficient is not None and abs(correlation_coefficient) >
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(seed) for seed in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, ...{trial_result}...}}")
        results.append(trial_result)

    if all(result["conjecture_holds"] for result in results):
        mean_value = sum(result["metric_value"] for result in results) / len(results)
        std_value = (sum((result["metric_value"] - mean_value) ** 2 for result in results) / len(r
        support_fraction = 1.0
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif any(not result["conjecture_holds"] for result in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conj
        print(f"RESULT: FALSIFIED counterexample=\"not supported\" first_failing_seed={first_faili
    else:
        print("RESULT: INCONCLUSIVE")
```

## 6. Per-seed results

*(no seed-level data — test crashed before producing TRIAL: lines)*

## 7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3b313135.py", line 115, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3b313135.py", line 61, in run_trial
    cnf = generate_cnf(n)
          ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3b313135.py", line 19, in generate_cnf
    return [[random.choice([-i, i]) for _ in range(n)] for _ in range(2 * n)]
                            ^
NameError: name 'i' is not defined. Did you mean: 'id'?
```

## 8. Critic adversarial review

**Critic verdict**: CHALLENGE

**Critic reasoning**:

> skipped: test crashed before producing data

## 9. Final verdict & safety rail

**Verdict**: INCONCLUSIVE

**Reasoning**:

> The test code crashed due to a NameError, which prevented it from producing any data or results. | next: Debug the test code to fix the NameError and rerun the test to verify the conjecture.

## 11. Audit log (LLM calls)

**Total LLM calls**: 10

| # | Phase | Provider | Model | Tokens in | out | Latency (ms) |
|---|---|---|---|---|---|---|
| 1 | propose | ollama_remote | glm4:latest | 0 | 0 | 21803 |
| 2 | propose | ollama_remote | glm4:latest | 0 | 0 | 17277 |
| 3 | preregistration | ollama_remote | glm4:latest | 0 | 0 | 9644 |
| 4 | novelty | ollama_remote | glm4:latest | 0 | 0 | 10459 |
| 5 | novelty | ollama_remote | glm4:latest | 0 | 0 | 22017 |
| 6 | test_gen | ollama_remote | qwen2.5-coder:7b | 0 | 0 | 25880 |
| 7 | test_gen | ollama_remote | qwen2.5-coder:7b | 0 | 0 | 22922 |
| 8 | test_gen | ollama_remote | qwen2.5-coder:7b | 0 | 0 | 19722 |
| 9 | test_gen | ollama_remote | qwen2.5-coder:7b | 0 | 0 | 13495 |
| 10 | judge | ollama_remote | glm4:latest | 0 | 0 | 12816 |

**Totals**: 0 input tokens, 0 output tokens, ~$0.0000 cost, 176035 ms total latency. Provider mix: {'ollama_remote': 10}

*(full prompt+response transcripts available in `research/audit/4ccef7b35a0b.jsonl`)*

## 12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/4ccef7b35a0b.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

**Sandbox file**: `research/pvsnp_sandbox/test_*.py` (per-cycle)
**Replay tarball**: `research/replay/4ccef7b35a0b.tar.gz` (if generated)